

B

1. (a) $\frac{10}{19}$

(b) $\frac{5+4}{19} = \frac{9}{19}$

(c) $\frac{9}{19} + \frac{9}{19} - \frac{4}{19} = \frac{14}{19}$

(d) $\frac{4}{9}$

(e) $P(\text{red} | \text{even}) = \frac{5}{9} \approx 0.5556$, but $P(\text{red}) = \frac{10}{19} \approx 0.5263$,

so they are not independent

2. (a) $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$$= 0.77 + 0.73 - 0.7$$

$$= \underline{0.8}$$

(b) $P(\text{neither } A \text{ nor } B) = P((A \text{ or } B)^c) = 1 - P(A \text{ or } B)$

$$= 1 - 0.8 = \underline{0.2}$$